Cosmic Muon Lab

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(Dated: October 4, 2024)

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I. INTRODUCTION

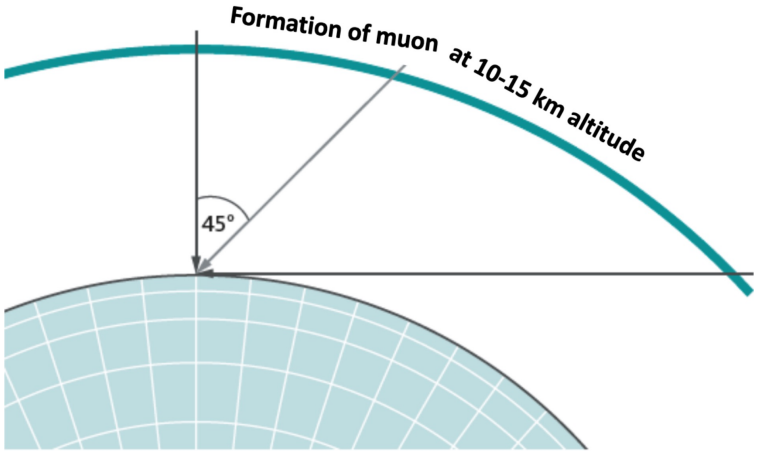


FIG. 1: Visualization of muon trajectory through the atmosphere. The trajectory of the muon becomes *significantly* longer as the angle becomes more horizontal. Image originally shown in lab manual [2].

The goal of this lab is to measure the rates of cosmic muons coming from space. These muons have a highly angular dependent distribution, as shown in Fig. 1. Muons/anti-muons that come from the cosmos

decay in the atmosphere decay into electrons/positrons,

$$\mu^{\pm} \rightarrow e^{\pm} + \bar{\nu}_e + \nu_\mu. \tag{1}$$

This process emits a level of light that can be detected using scintillator telescopes with photomultiplier tubes (PMTs). Using two such scintillators on a rotating axis allows for the light from this decay process to be detected and recorded.

The goal for this lab is to determine the muon decay rate. This can be determined by first calibrating the equipment (list shown in Sec. II A) used in the lab. Once calibrated, the rates of detection can be converted to time, and an exponential curve can be measured. The curve should take on the following form,

$$R(t) = R_0\, e^{-t/\tau}, \tag{2}$$

where R_0 is the muon detection rate at time $t = 0s$ and τ is the mean lifetime of the muon.

II. PROCEDURE AND EQUIPMENT

A. Equipment Description

The equipment used in this lab includes:

1. Two scintillator telescopes
2. Four photomultipliers
3. Rotating apparatus
4. Electrical equipment:
   (a) Four amplifiers
   (b) Four discriminators
   (c) Three coincidence units
   (d) Two delays
   (e) Potentiometer
   (f) Necessary wiring
5. Time-Analog-Counter (TAC)
6. Power supply (capable of delivering 1320 volts)
7. Oscilloscope
8. Timer or clock

This equipment is all accessible in the Edwards Accelerator Laboratory and open for use to the students in the graduate nuclear lab course. The electronic equipment should be setup according to the diagram in Fig. 2. Once setup and correctly wired, the power supply should be set to deliver 1320 volts to the scintillators.

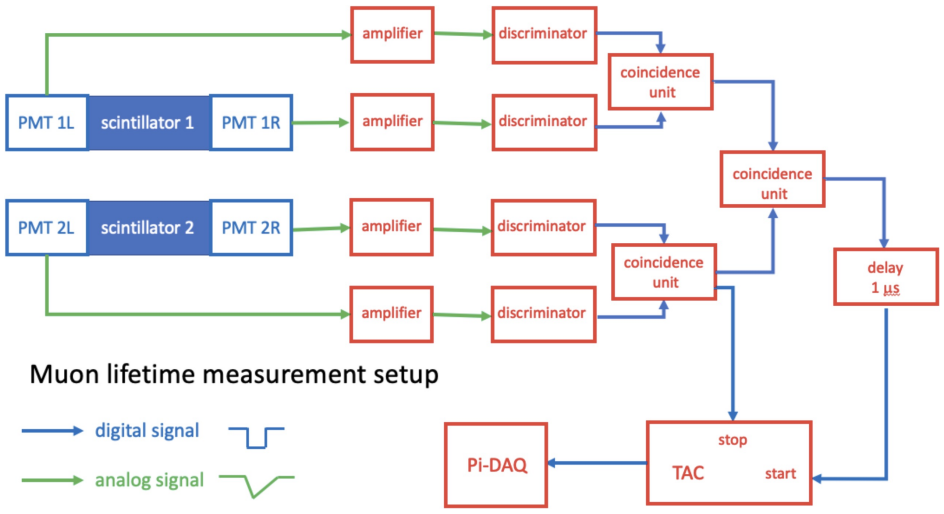


FIG. 2: Setup for electronics for the lab. When calibrating the equipment, add an extra delay between the lower coincidence unit and the TAC. This allows for the newly added delay to be set between $1\mu s$ and $10\mu s$, or as high as is allowed by the equipment. Image originally shown in lab manual [2].

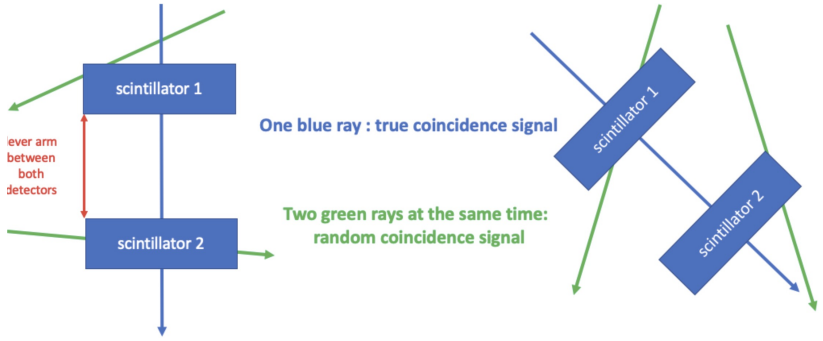


FIG. 3: Diagram of scintillator setup. As seen, the scintillator telescopes pivot around a common fulcrum. This allows form muon detection along an azimuthal angle. This allows for both ease of detection of muons without drastic changes to the laboratory setup, as well as the detection of the background when roughly zero muons are expected to be detected (at the horizontal). Image originally shown in lab manual [2].

The scintillator telescopes should be set up as shown in Fig. 3, allowing for ease of rotation of the scintillators.

B. Equipment Calibration

It is crucial for the accuracy and reputability of the experiment for the equipment to be properly calibrated. Such calibration requires the systematic testing of the TAC and the use of an oscilloscope for setting the time delay in the TAC to specified values of $1\mu s$ to $8\mu s$, in steps of $1\mu s$.

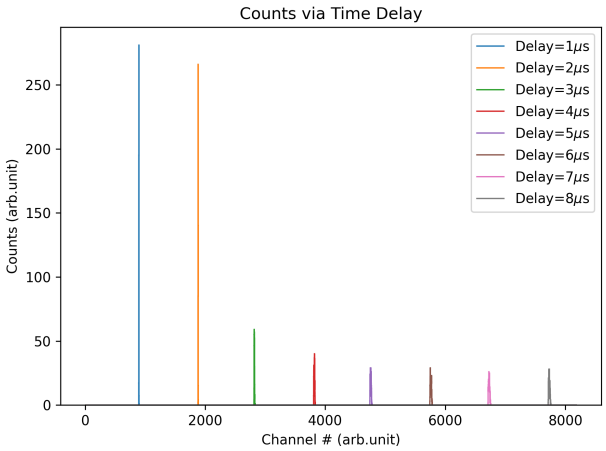


FIG. 4: The raw data for the time calibration. Each peak is associated with a two minute long data acquisition time. The total number of counts per peak is approximately constant. The shape of each peak is approximately Gaussian.

The result of the calibration for this lab is shown in Fig. 4. Each data point shows the number of counts for a two-minute run at each time interval. This choice was made to keep the most consistency in the lab. This also allowed for any glaring inconsistencies from the calibration portion of the experiment to be detected. These peaks were used to calibrate the detector, using the known knowledge of both the discriminator threshold of 8mV and timing of the additional delay. The mean value of each Gaussian signal was used to find the conversion between channel/bin and time delay.

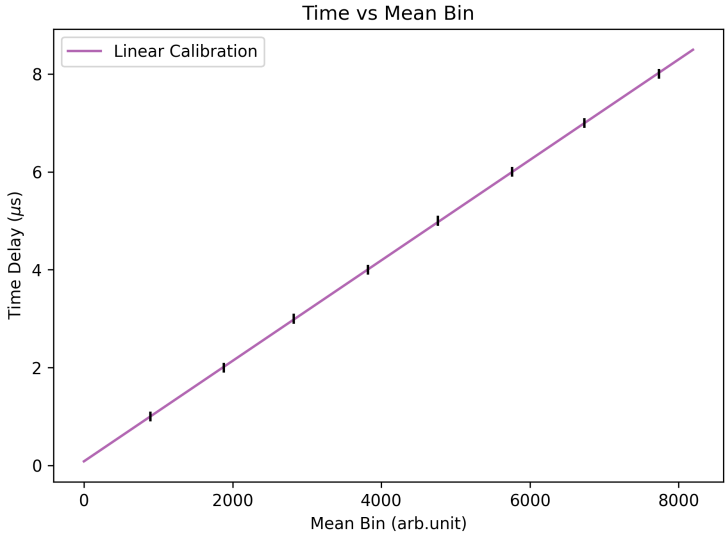


FIG. 5: Time calibration for the TAC. The locations of each bin was determined through a Gaussian fit. The error bars on each point is $1\mu s$ long, to account for potential errors in setting the time delay.

TABLE I: Data values from Fig. 5. Peak locations have been rounded off to a single decimal place. The peak location was determined by fitting each peak with a Gaussian and taking the location of the Gaussian's peak.

Time Delay	$1\mu s$	$2\mu s$	$3\mu s$	$4\mu s$	$5\mu s$	$6\mu s$	$7\mu s$	$8\mu s$
Peak Location	896.6	1883.2	2820.1	3819.1	4759.6	5760.9	6730.9	7734.0

This allows for us to convert the number of counts in each bin/channel to the time delay. The eight spikes in Fig. 5 show the eight time delays used as plotted with the time associated with each peak. The vertical error bars shown in the plot are $0.1\mu s$ long and are intended to capture human error in setting the time delay.

Each of the PMTs be set to the correct voltage for the correct reading to occur. This voltage is found by scanning across a range of voltages allowed by the PMTs. Readings are taken for how many counts are found within a specific time frame, five seconds is sufficient. The ideal voltage settings is found by locating the point at which the counts remain approximately constant within a globally decreasing tend in the data. Each PMT is set to its corresponding voltage for the data acquisition portion of the lab.

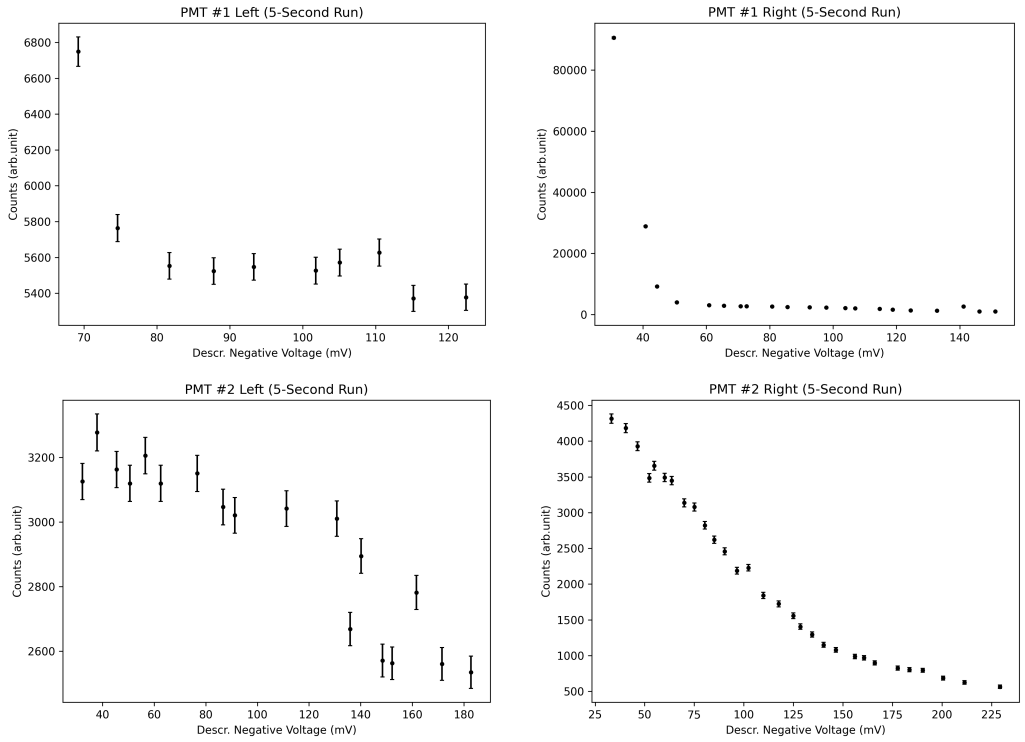


FIG. 6: Plots for each of the PMTs in the setup. Each point represents the counts for a five second data run. Due to the shorter runtime and the Poisson distribution given by counting statistics, the points are somewhat imprecise. The error bars on the measurements are of size $\sqrt{N_{\text{counts}}}$.

This process was done in the lab and is visualized in Fig. 6. The used for this lab are shown in table II. This allows for the PMTs to operate as expected when proceeding to the next part of the lab.

TABLE II: PMT voltage settings as determined by the process mentioned in Sec. II B.

PMT Location	#1 left	#1 Right	#2 Left	#2 Right
PMT Voltage	−96.6 mV	−97.7 mV	−101.5 mV	−183.6 mV

After the voltages were found the coincidence rates were found for each location in the tube. This was done by taking one minute time runs, resulting in the values in Table III.

TABLE III: Coincidence rates for the PMTs individually, and the scintillator telescopes as a whole. As more detectors are included in the coincidence rate, the lower the count rates are unsurprisingly lower. The rates from "Both Scintillator Telescopes" are measurements with the least amount of background.

PMT Location	Counts/Second
Top Left PMT	1136.00
Top Right PMT	454.98
Bottom Left PMT	661.63
Bottom Right PMT	151.93
Top Scintillator Telescope	92.10
Bottom Scintillator Telescope	62.77
Both Scintillator Telescopes	4.22

III. VERIFICATION OF MUON ANGULAR DEPENDENCE

Using the calibrated equipment is used to detect cosmic muons at different azimuthal angles. This was done by setting the voltages in each PMT to the values obtained in Sec. II B. Then the TAC was used to count the number of muons and background at angles of 90° (vertical) to 0° (horizontal) in steps of 10°. The choice of 10° steps was chosen by using a geometric argument of the precision of the scintillator setup. Data was taken in one minute intervals for each angle. This process was repeated three times, and the resulting data was combined to form Figs. 7 and 8.

As mentioned before, the measured counts follow a Poisson distribution. This fact leads the error bars in the y-direction to scale is $\sqrt{N_{\text{counts}}}$. Error bars were added in the x-direction of two degrees to account for possible imprecise setting of the telescope angle. Data for determining the angular dependence of cosmic muon rates was taken in one minute time intervals, and is shown in Fig. 7.

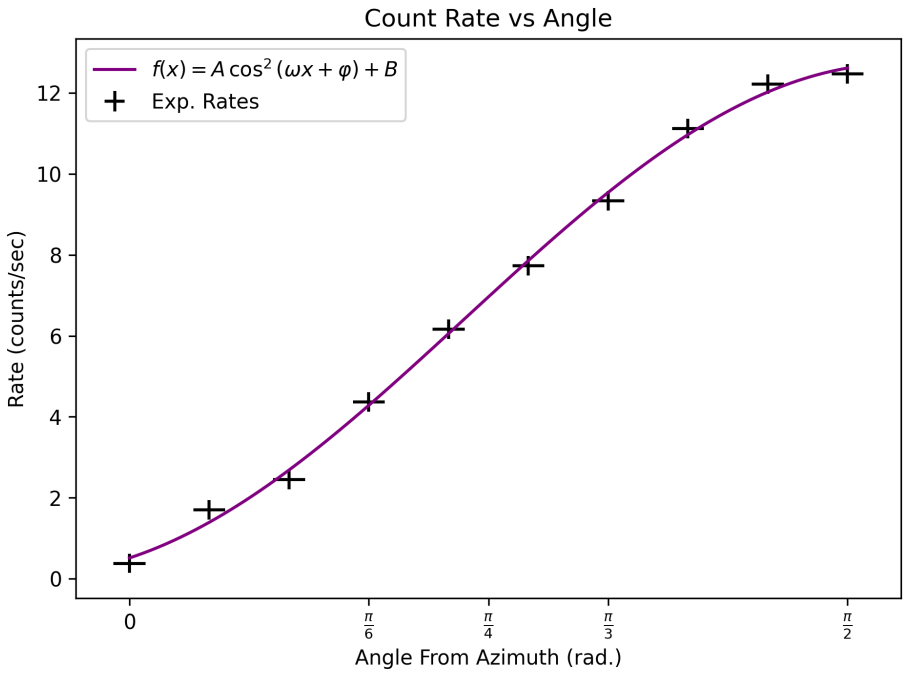


FIG. 7: Results plotted against angle. 0 radians represents the vertical, and $\frac{\pi}{2}$ represents the horizontal. Horizontal error bars of two degrees have been added to account for inconsistencies in setting the angle.

The information from Fig. 7 is then plotted against $\cos^2\theta$ rather than just θ so that a linear fit could be used. The fit in Fig. 8 is the same fit that is mentioned in the course notes,

$$f(x) = \alpha x + \beta \tag{3}$$

$$\Delta = DB - A^2 \qquad \alpha = \frac{EB - CA}{\Delta} \qquad \beta = \frac{DC - EA}{\Delta} \tag{4}$$

$$\tag{5}$$

$$\mathrm{cov}(\alpha, \beta) = -\frac{A}{\Delta} \qquad \sigma_\alpha^2 = \frac{B}{\Delta} \qquad \sigma_\beta^2 = \frac{D}{\Delta}, \tag{6}$$

where,

$$A = \sum_i \frac{x_i}{\sigma_i^2} \qquad B = \sum_i \frac{1}{\sigma_i^2} \qquad C = \sum_i \frac{y_i}{\sigma_i^2} \tag{7}$$

$$D = \sum_i \frac{x_i^2}{\sigma_i^2} \qquad E = \sum_i \frac{x_i\, y_i}{\sigma_i^2} \qquad F = \sum_i \frac{y_i^2}{\sigma_i^2}. \tag{8}$$

This type of fit allows for a straightforward error on the fit to be found, as well as the covariance.

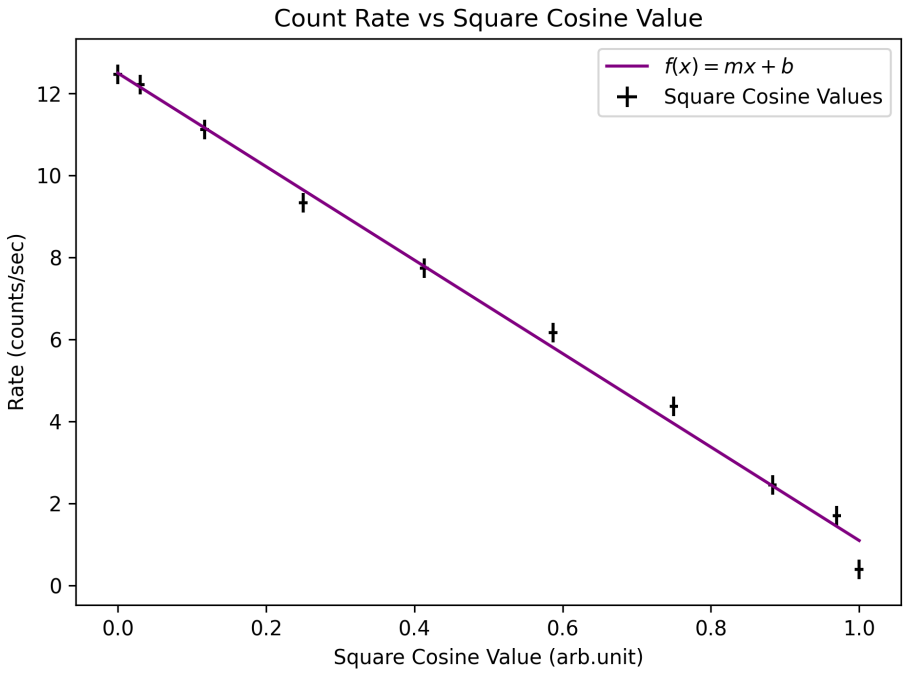


FIG. 8: Results plotted as line. Horizontal error bars of two degrees have been added to account for inconsistencies in setting the angle.

The results from fitting the combined set of data is shown in Table IV.

Fit Parameter	Fit Value
$\alpha \pm \sigma_\alpha$	-11.40 ± 0.20
$\beta \pm \sigma_\beta$	12.50 ± 0.13
$\mathrm{cov}(\alpha, \beta)$	-0.02

TABLE IV: Calculated values from Equations (3) to (6). The fit that results from these values are shown in Fig. 8. All values are rounded to two decimal places.

IV. ACKLOWLEDGEMENTS

We thank Prof. Paul King for all the help extended in performing and understanding of the experiment. I also acknowledge and thank Cade Saugier for being my lab partner for this experiment.

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[King] King, P. Physics 6751 course website.
[2] Lab Manual (2024). *PHYS6751: Nuclear Graduate Lab Lab manual v24.5.*